

Day 16 Plan

Maths: Geometry · Reasoning: Mirror & Water Images, Paper Folding, Embedded Figures · Science: Biology 3: Human Physiology (Life Processes)

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 15 recall	15 min	10 questions from Day 15, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	7 lessons with quick checks (≈65 min) → 20-question practice set → Testbook PYQs: human body, life processes (A tier)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Environment — conventions, national parks, biosphere reserves — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 15 recall (answers on the last page)

#	Question	Your answer
1	pH of human blood?	
2	Formula of washing soda?	
3	Gas given off when a metal reacts with dilute acid?	
4	Isotope used in cancer treatment?	
5	Bronze is made of?	
6	Main component of LPG?	
7	Coal with the highest carbon content?	
8	Lime water turns milky because of?	
9	Solder is made of?	
10	Maths: $x + 1/x = 5 \rightarrow x^2 + 1/x^2 = ?$	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Geometry

Core idea

RRB geometry is about **angle facts** and a handful of circle and triangle theorems — not long proofs. Draw a rough figure for every question and write the angles on it. Most questions need one theorem plus one subtraction.

Formulas

Concept	Formula / rule
Triangle	Angles sum to 180° · exterior angle = sum of the two interior opposite angles
Polygon (n sides)	Interior sum $(n - 2) \times 180^\circ$ · each exterior angle of a regular polygon = $360^\circ/n$ · diagonals $n(n - 3)/2$
Parallel lines	Alternate angles equal · corresponding equal · co-interior angles add to 180°
Similar triangles	Ratio of areas = (ratio of sides) ²
Centroid	Divides each median 2 : 1 from the vertex
Equilateral triangle (side a)	Circumradius $a/\sqrt{3}$ · inradius $a/(2\sqrt{3})$ · all centres coincide
Circle angles	Angle at the centre = $2 \times$ angle at the circumference · angle in a semicircle = 90° · angles in the same segment are equal
Cyclic quadrilateral	Opposite angles add to 180°
Tangents	Tangent $\perp$ radius · two tangents from a point are equal · tangent length = $\sqrt{d^2 - r^2}$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Angle ratio	Share 180°	Angles 2 : 3 : 4 $\rightarrow 180 \div 9 = 20 \rightarrow$ 40°, 60°, 80°
2. Exterior angle	Sum of opposite interiors	Exterior 110° , one opposite angle $45^\circ \rightarrow$ other 65°
3. Regular polygon	$360 \div$ exterior	Each exterior angle $40^\circ \rightarrow$ 9 sides ; interior angle 140°
4. Diagonals	$n(n - 3)/2$	Hexagon $\rightarrow 6 \times 3 \div 2 =$ 9
5. Interior sum	$(n - 2) \times 180$	Octagon $\rightarrow$ 1080°
6. Similar triangles	Square the ratio	Sides 3 : 5 $\rightarrow$ areas 9 : 25
7. Centre vs circumference	Halve	Arc subtends 100° at the centre $\rightarrow$ 50° at the circumference
8. Cyclic quadrilateral	$180 -$ angle	One angle $70^\circ \rightarrow$ opposite angle 110°
9. Tangent length	$\sqrt{d^2 - r^2}$	Point 13 cm from the centre, radius 5 $\rightarrow$ 12 cm
10. Equilateral circumradius	$a/\sqrt{3}$	Side $6\sqrt{3} \rightarrow$ 6

Traps

- Interior vs exterior angle — read which one is given.
- The centroid splits the median 2 : 1 **from the vertex**.
- Area ratio of similar figures is the **square** of the side ratio, not the ratio itself.

B · REASONING — Mirror & Water Images, Paper Folding, Embedded Figures

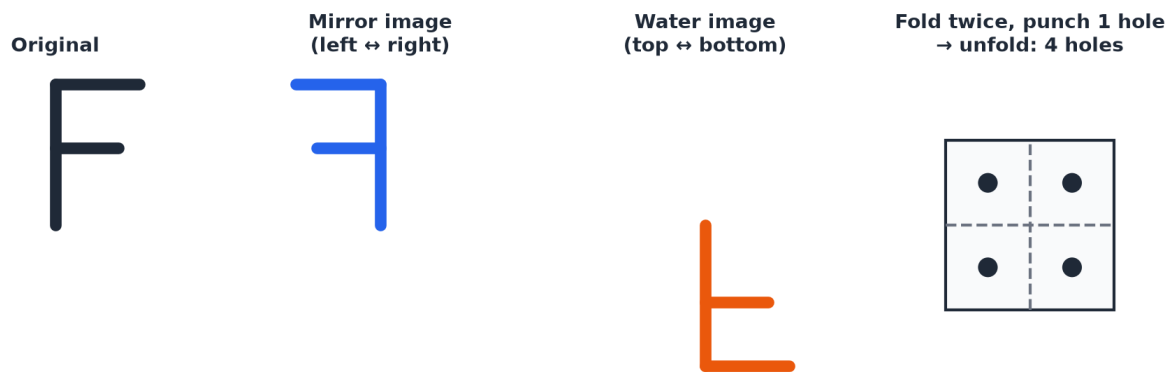


Image	What flips	Letters that look the same
Mirror (vertical mirror)	Left ↔ right; top stays top	A H I M O T U V W X Y
Water (below the object)	Top ↔ bottom; left stays left	B C D E H I K O X

Clock in a mirror: mirror time = **11:60 – actual time** (e.g. 4:20 → 7:40). If the answer is above 12, subtract 12.

Paper folding: unfold one step at a time; every hole is reflected across the fold line, so each unfold doubles the holes (fold twice, punch once → 4 holes). **Embedded figures:** find a distinctive corner or angle of the small figure and look for that first.

Patterns, with solved examples

Pattern	Example	Solution
Mirror clock	A clock shows 4:20. Time in the mirror?	$11:60 - 4:20 = \mathbf{7:40}$
Mirror clock	A clock shows 2:45. Time in the mirror?	$11:60 - 2:45 = \mathbf{9:15}$
Mirror letters	Which looks the same in a vertical mirror: A, B, K, P?	A
Water word	Which word looks the same in water: DICE or RAIL?	DICE (D, I, C, E are symmetric top-to-bottom)
Paper folding	A square sheet is folded in half twice and one hole is punched. Holes when opened?	4
Paper folding	Folded in half three times, one hole punched?	8

Speed rule

For figure questions, eliminate options using one feature at a time (a corner, a dot, a shaded part) instead of comparing whole figures.

C · SCIENCE — Biology 3: Human Physiology (Life Processes)

7 lessons on how the human body works. Human physiology was 11 of 210 science questions (R), and the 'largest/smallest/normal value' facts in Lesson 7 come up across papers.

Lesson 1 · Respiration and breathing (about 9 min)

Type	Equation	Where
Aerobic	Glucose + O ₂ → CO ₂ + H ₂ O + a lot of energy	Mitochondria, most cells
Anaerobic (yeast)	Glucose → ethanol + CO ₂ + a little energy	Fermentation — bread, alcohol
Anaerobic (muscles)	Glucose → lactic acid + a little energy	Hard exercise — causes cramps

Air path: nostrils → pharynx → **larynx** (voice box) → **trachea** (windpipe, kept open by C-shaped cartilage rings) → bronchi → bronchioles → **alveoli**, tiny air sacs with a huge surface area where O₂ and CO₂ are exchanged. The **diaphragm** moves down to breathe in.

Haemoglobin in red blood cells carries oxygen.

Exam trap

Breathing is the physical movement of air; **respiration** is the chemical release of energy from food inside cells.

Quick check 1 — answer out loud before moving on (answers at the end)

a) Product of anaerobic respiration in muscles? b) Where in the lungs are gases exchanged?

Lesson 2 · Blood (about 10 min)

Part	Facts
Plasma (≈ 55%)	Pale yellow fluid; carries food, CO ₂ , hormones, waste
Red blood cells (RBC)	Carry O ₂ (haemoglobin, contains iron) · no nucleus in mammals · live ≈ 120 days · made in bone marrow · old ones destroyed in the spleen ('graveyard of RBCs')
White blood cells (WBC)	Fight infection (immunity) · leukaemia = cancer of WBC-making tissue
Platelets	Help blood clot · fall sharply in dengue

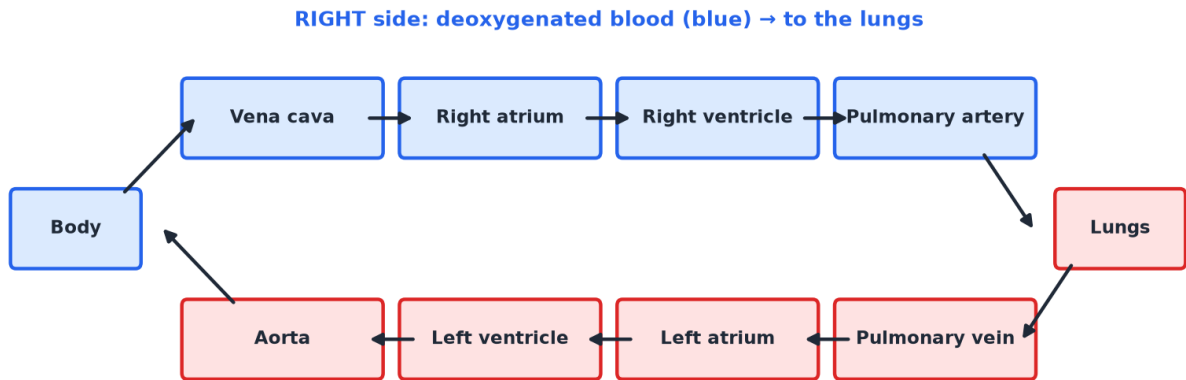
Blood group	Can donate to	Can receive from
O–	Everyone — universal donor	O– only
AB+	AB+ only	Everyone — universal recipient
A	A, AB	A, O
B	B, AB	B, O

Blood groups were discovered by **Karl Landsteiner**. Normal blood pressure ≈ **120/80 mm Hg** (sphygmomanometer). An adult has about 5–6 litres of blood. Clotting needs **vitamin K**; **haemophilia** is a genetic disorder in which blood doesn't clot. **Lymph** is colourless, carries digested fat and fights infection.

Quick check 2 — answer out loud before moving on (answers at the end)

a) Universal donor blood group? b) Lifespan of a red blood cell?

Lesson 3 · Heart and circulation (about 9 min)



LEFT side: oxygenated blood (red) → to the body · blood passes the heart twice = double circulation

The human heart has **4 chambers** — 2 atria (receive) and 2 ventricles (pump). Valves stop backflow. Blood passes through the heart twice in one round: **double circulation**. Heart rate ≈ **72 per minute**; the beat starts at the **SA node** (natural pacemaker).

Vessel	Carries	Exception
Artery	Blood away from the heart; thick walls, high pressure	Pulmonary artery carries deoxygenated blood
Vein	Blood to the heart; valves, thin walls	Pulmonary vein carries oxygenated blood
Capillary	Thin vessels where exchange happens	—

Heart chambers in other animals: fish **2** · amphibians and most reptiles **3** · crocodile, birds, mammals **4**.

Quick check 3 — answer out loud before moving on (answers at the end)

a) Which artery carries deoxygenated blood? b) How many chambers does a fish's heart have?

Lesson 4 · Excretion (about 7 min)

Two bean-shaped **kidneys** filter blood. Path: kidney → **ureter** → urinary bladder → urethra. The functional unit is the **nephron**: blood is filtered in the glomerulus (inside Bowman's capsule), then useful substances are reabsorbed in the tubule.

Urine is water, **urea** and salts. Urea is made in the **liver**. If kidneys fail, **dialysis** (artificial kidney) cleans the blood. Kidney stones are mostly calcium oxalate. Other excretory routes: lungs (CO₂), skin (sweat). Plants excrete through stomata, falling leaves, resins and gums.

Quick check 4 — answer out loud before moving on (answers at the end)

a) Functional unit of the kidney? b) Where is urea made?

Lesson 5 • Nervous system (about 8 min)

Part	Controls
Cerebrum (largest part)	Thinking, memory, intelligence, voluntary actions, senses
Cerebellum	Balance , posture, coordination of movements
Medulla oblongata	Involuntary actions: heartbeat, breathing, blood pressure, vomiting, salivation
Hypothalamus	Body temperature, hunger, thirst; controls the pituitary
Spinal cord	Reflex actions (pulling your hand from a hot object)

A **neuron** carries impulses: dendrites → cell body → axon; the gap between neurons is a **synapse**. The brain is protected by the skull, membranes (meninges) and cerebrospinal fluid.

Quick check 5 — answer out loud before moving on (answers at the end)

a) Which part of the brain keeps balance? b) Reflex actions are controlled by?

Lesson 6 • Hormones (endocrine glands) (about 8 min)

Gland	Hormone	Effect / disorder
Pituitary (master gland)	Growth hormone, others	Too little → dwarfism; too much → gigantism
Thyroid	Thyroxine (needs iodine)	Metabolism • goitre
Parathyroid	Parathormone	Blood calcium level
Adrenal	Adrenaline	'Fight or flight' emergency hormone
Pancreas (islets of Langerhans)	Insulin , glucagon	Insulin lowers blood sugar • lack → diabetes
Testes • ovaries	Testosterone • oestrogen, progesterone	Puberty changes
Pineal • thymus	Melatonin • thymosin	Sleep cycle • immunity (T-cells)

Insulin was discovered by **Banting and Best** (1921). Endocrine glands are ductless — they pour hormones straight into the blood.

Quick check 6 — answer out loud before moving on (answers at the end)

a) Which gland is called the master gland? b) Lack of which hormone causes diabetes?

Lesson 7 · Body records, senses and reproduction (about 9 min)

Fact	Answer	Fact	Answer
Bones in an adult	206	Longest bone	Femur (thigh)
Smallest bone	Stapes (in the ear)	Largest organ	Skin
Largest gland	Liver	Hardest substance	Tooth enamel
Adult teeth	32	Normal body temperature	37 °C (98.6 °F)
Ear bones	Malleus, incus, stapes	Image forms on	Retina
Rods · cones	Dim light · colour	Transplanted eye part	Cornea

Reproduction: fertilisation happens in the **fallopian tube (oviduct)**; the embryo implants in the uterus and is fed through the **placenta**; pregnancy lasts about 9 months ($\approx$ 280 days). **Sex of the child** is decided by the **father**: an X sperm gives a girl (XX), a Y sperm a boy (XY).

Quick check 7 — answer out loud before moving on (answers at the end)

a) Smallest bone in the body? b) Which parent's chromosome decides the sex of the child?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Respiration	Aerobic → $\text{CO}_2 + \text{H}_2\text{O}$ · yeast → ethanol + CO_2 · muscles → lactic acid · alveoli exchange gases · larynx voice box
Blood	RBC 120 days, bone marrow, spleen graveyard · platelets clot (dengue ↓) · O— universal donor · AB+ universal recipient · Landsteiner
Heart	4 chambers · double circulation · pulmonary artery deoxygenated, pulmonary vein oxygenated · 72/min · SA node · fish 2 chambers
Kidney	Nephron · urea made in liver · dialysis · stones calcium oxalate
Brain	Cerebrum thinking · cerebellum balance · medulla involuntary · spinal cord reflexes
Hormones	Pituitary master · thyroxine iodine · adrenaline emergency · insulin lowers sugar (Banting & Best)
Records	206 bones · femur longest · stapes smallest · skin largest organ · liver largest gland · enamel hardest · 37 °C
Reproduction	Fertilisation in fallopian tube · placenta · father decides sex (XY)

Science practice set — 20 questions · target 6 minutes

#	Question
1	Anaerobic respiration in muscles produces: (a) ethanol (b) acetic acid (c) lactic acid (d) citric acid
2	Exchange of gases in the lungs takes place in the: (a) trachea (b) bronchi (c) larynx (d) alveoli
3	The voice box is the: (a) pharynx (b) trachea (c) larynx (d) epiglottis
4	The lifespan of a human red blood cell is about: (a) 30 days (b) 120 days (c) 60 days (d) 365 days
5	The 'graveyard of RBCs' is the: (a) spleen (b) liver (c) kidney (d) heart
6	The universal donor blood group is: (a) AB positive (b) A positive (c) O negative (d) B negative
7	Blood groups were discovered by: (a) William Harvey (b) Louis Pasteur (c) Robert Koch (d) Karl Landsteiner
8	Which blood cells help in clotting? (a) Platelets (b) RBCs (c) WBCs (d) Plasma cells
9	Which blood vessel carries deoxygenated blood from the heart to the lungs? (a) Pulmonary vein (b) Aorta (c) Vena cava (d) Pulmonary artery
10	A fish's heart has: (a) 3 chambers (b) 4 chambers (c) 1 chamber (d) 2 chambers
11	The natural pacemaker of the heart is the: (a) AV node (b) aorta (c) left ventricle (d) SA node
12	The functional unit of the kidney is the: (a) neuron (b) nephron (c) alveolus (d) villus
13	Urea is made in the: (a) kidney (b) bladder (c) pancreas (d) liver
14	Balance and posture are controlled by the: (a) cerebrum (b) cerebellum (c) medulla oblongata (d) hypothalamus
15	Heartbeat and breathing are controlled by the: (a) medulla oblongata (b) cerebrum (c) cerebellum (d) spinal cord
16	The 'master gland' is the: (a) thyroid (b) adrenal (c) pituitary (d) pancreas
17	Insulin is secreted by the: (a) liver (b) thyroid (c) pancreas (d) adrenal gland
18	The 'emergency' hormone is: (a) insulin (b) thyroxine (c) adrenaline (d) melatonin
19	The longest bone in the human body is the: (a) tibia (b) humerus (c) femur (d) fibula
20	Fertilisation in humans takes place in the: (a) uterus (b) ovary (c) fallopian tube (d) cervix

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	c	Lesson 1
2	d	Lesson 1
3	c	Lesson 1
4	b	Lesson 2
5	a	Lesson 2
6	c	Lesson 2
7	d	Lesson 2
8	a	Lesson 2
9	d	Lesson 3
10	d	Lesson 3
11	d	Lesson 3
12	b	Lesson 4
13	d	Lesson 4
14	b	Lesson 5
15	a	Lesson 5
16	c	Lesson 6
17	c	Lesson 6
18	c	Lesson 6
19	c	Lesson 7
20	c	Lesson 7

Quick checks

QC	Answers
1	a) Lactic acid b) Alveoli
2	a) O negative b) About 120 days
3	a) Pulmonary artery b) 2
4	a) Nephron b) Liver
5	a) Cerebellum b) Spinal cord
6	a) Pituitary b) Insulin
7	a) Stapes (in the ear) b) The father's

Day 15 recall (Block 0)

#	Answer	#	Answer
1	≈ 7.4 (7.35–7.45)	6	Butane
2	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	7	Anthracite
3	Hydrogen	8	CO_2
4	Cobalt-60	9	Lead + tin
5	Copper + tin	10	23

Send me tonight

Day 15 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.